

Santiago A. Ruiz Diaz

OT/ICS Systems Engineer | Building Automation & Controls | Security+ · CySA+ · PenTest+ | 15+ Years Critical Infrastructure

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PROFESSIONAL SUMMARY

Operational Technology (OT) and Industrial Control Systems (ICS) professional with 15+ years of hands-on experience maintaining and troubleshooting Building Management Systems (BMS), SCADA-like controllers, PLC-driven automation (GCL+), and BACnet networks across a live, network-connected multi-building campus.

That field-level ownership — system monitoring, PLC logic repair, configuration changes, and access control administration across HVAC, boiler, and electrical infrastructure — is paired with a security engineering skill set most controls engineers don't have: vulnerability management, threat hunting, and endpoint detection across a ~200-server enterprise environment.

Currently working as Security Engineer I at LOGN Pacific, delivering hands-on projects in Tenable vulnerability management, Microsoft Sentinel, and Defender for Endpoint — while completing my B.S. in Cybersecurity & Information Assurance at Western Governors University. Pursuing a Master's degree in Cybersecurity with enrollment planned for mid-2026.

CORE COMPETENCIES

OT/ICS Operations · Building Management Systems (BMS) · SCADA · PLC Programming & Troubleshooting · GCL+ · BACnet · HVAC & Electrical Infrastructure · IEC 62443 · NIST 800-82 · IT/OT Convergence · Network Segmentation · Purdue Model · System Monitoring & Fault Diagnosis · Access Control · Privileged Access Management · Change Management · Risk Assessment · SIEM · Threat Hunting · Incident Response · Vulnerability Assessment · EDR · Zero Trust · System Hardening · Python · PowerShell · Bash · KQL · Microsoft Sentinel · Microsoft Defender for Endpoint · Tenable / Nessus · Azure Security · MITRE ATT&CK · DISA STIG · NIST CSF · NIST 800-53

CERTIFICATIONS

CompTIA PenTest+ — CompTIA · Issued: Mar 2026 · Expires: Mar 2029

CompTIA CySA+ (ce) — CompTIA · Issued: Aug 2025 · Expires: Aug 2031

CompTIA Security+ (ce) — CompTIA · Issued: Jul 2025 · Expires: Jul 2028

CompTIA Network+ (ce) — CompTIA · Issued: Jul 2025 · Expires: Jul 2028

CompTIA A+ (ce) — CompTIA · Issued: May 2025 · Expires: Aug 2031

CompTIA Secure Infrastructure Specialist (CSIS) — CompTIA · Issued: Jul 2025 · Expires: Jul 2028

CompTIA IT Operations Specialist (CIOS) — CompTIA · Issued: Jul 2025 · Expires: Jul 2028

Linux Essentials Certificate — LPI · Issued: May 2025

ITIL Foundation — AXELOS

EXPERIENCE

County College of Morris

HVAC & Building Automation Systems Specialist | OT/ICS Operations · November 2010 – Present (15+ years)

Technical owner of critical building infrastructure across a network-connected, multi-building campus — operating and maintaining OT/ICS systems, SCADA-like controllers, BACnet-connected devices, and PLC-driven building automation across HVAC, boilers, and electrical infrastructure.

PLC Scripting & System Maintenance

- Maintained, troubleshot, and repaired GCL+ PLC scripts across SCADA-like controllers and building automation systems — diagnosing and resolving logic errors and escalating out-of-scope issues to vendor/engineering teams.

- Made configuration changes and system adjustments via BMS client terminal to maintain operational integrity across HVAC, boilers, and electrical infrastructure — minimizing downtime and unplanned outages.

System Monitoring & Incident Escalation

- Monitored operational integrity of networked control systems across 15+ campus buildings, identifying anomalous behavior and system faults across BACnet-connected devices and electronic controls.
- Identified and escalated potential risks and unresolvable issues to appropriate teams, supporting system reliability and continuity of campus operations.
- Applied IT/OT convergence and network segmentation principles consistent with IEC 62443 and NIST 800-82 to isolate and protect building automation systems from the broader campus network.
- Collaborated with leadership to align OT operations with organizational change management and security policies.

Access Control & Identity Management

- Administered access control for building automation systems via BMS client terminal — provisioning, modifying, and revoking user permissions using least-privilege and privileged access management (PAM) principles.
- Conducted periodic access reviews to ensure authorization aligned with current roles and responsibilities.

LOGN Pacific

Security Engineer I · March 2026 – Present

Security engineering role applying an OT-informed operational lens to threat detection, vulnerability management, and incident investigation across Microsoft Azure, Microsoft Sentinel, Defender for Endpoint (EDR), and Tenable.

Vulnerability Management Program (Tenable)

- Implemented a full VM lifecycle across ~200-server enterprise environment (Windows Server 2025 & Linux) using Tenable — authenticated and unauthenticated scans with DISA STIG scan templates.
- Prioritized findings by CVSS score, asset criticality, and exposure: NTLMv1 disable, SMB signing enforcement, TLS 1.0/1.1 deprecation, and weak cipher remediation.
- Built PowerShell scripts for automated Windows server remediation and Bash scripts for Linux server remediation at scale.
- Facilitated CAB meetings, drafted VM policy aligned to NIST CSF and NIST 800-53, and drove executive sign-off and server team buy-in.

Signal After The Noise — Post-Intrusion Threat Hunt (Microsoft Sentinel)

- Reconstructed a full adversary attack chain in Microsoft Sentinel across a simulated enterprise environment using SOC threat hunting methodology.
- Authored 20+ KQL queries across 6 MDE log tables: DeviceLogonEvents, DeviceProcessEvents, DeviceFileEvents, DeviceNetworkEvents, DeviceRegistryEvents, and DeviceEvents.
- Mapped full kill chain: T1078 (credential reuse) → T1021 (RDP lateral movement) → T1547.001 (persistence) → T1071.001 (C2 beaconing) → T1562.001 (AMSI evasion) → T1036 (LOLBin masquerade) → T1003.001 (LSASS credential dump).
- Decoded Base64-encoded PowerShell payloads using KQL's `base64_decode_tostring()` to expose Cloudflare-fronted C2 infrastructure.
- Produced full IOC table and MITRE ATT&CK-mapped incident report.

TOR Network Detection (Microsoft Defender for Endpoint / EDR)

- Confirmed unauthorized TOR browser usage on a corporate Azure Windows 11 endpoint via KQL across DeviceFileEvents, DeviceProcessEvents, and DeviceNetworkEvents.

- Detected full TOR circuit — tor.exe to external relay on port 9001, firefox.exe from non-standard Desktop path to evade registry traces, suspicious file created and deleted to conceal activity.
- Mapped to MITRE T1090.003 (proxy), T1204 (user execution), T1036 (masquerading). Delivered incident report with device isolation, HR/legal escalation, and perimeter block of TOR ports 9001–9151.

PROJECTS

Full documentation: github.com/Santi4g0RD

Vulnerability Management Program

Tools: Tenable, PowerShell, Bash | Full VM lifecycle across ~200-server enterprise — scanning, CVSS prioritization, remediation scripting on Windows and Linux, VM policy drafting, and stakeholder coordination.

Signal After The Noise — Post-Intrusion Threat Hunt

Tools: Microsoft Sentinel, KQL | Reconstructed full adversary kill chain across 6 MDE log tables using 20+ KQL queries; mapped to MITRE ATT&CK framework end-to-end.

TOR Network Detection

Tools: Microsoft Defender for Endpoint (EDR), KQL | Detected and investigated unauthorized TOR usage on corporate endpoint; produced full IOC documentation and incident report with remediation recommendations.

EDUCATION

B.S. Cybersecurity & Information Assurance (BSCSIA) — Western Governors University · Mar 2025 – Mar 2026

Master's Degree in Cybersecurity — Enrollment target: July–August 2026

Coursework — Electronics & Electrical Engineering Technology — Thomas Edison State University · 2014–2017

LANGUAGES

English — Native / Bilingual | Spanish — Native / Bilingual